SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	**Question**	**Bloom's Level**
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

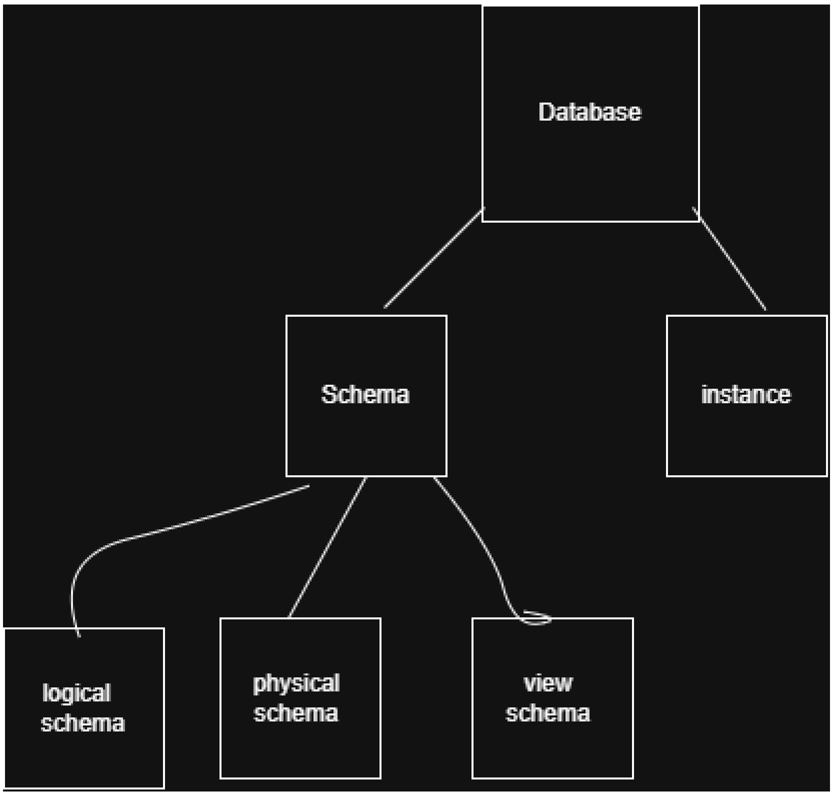
Criteria	**Weightag e**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconception s	Poor or incorrect understandin g
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organizatio n	20%	Correctly applies concepts to solve the question/proble m	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandab l e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Question 1

Explain Schema and Instance with examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

Tool Used: Draw.io (diagrams.net)

Concept Map:-



1. Schema

Definition :-

A **Schema** is the blueprint or overall design of a database. It defines the structure of the database, including tables, attributes (columns), data types, relationships, and constraints. The schema is created during database design and changes very rarely.

Features

- Defines the structure of the database.
- Includes tables, columns, and relationships.
- Does not contain actual data.

- Created by the database designer or DBA.

Example

Suppose we have a **Student** table.

Schema :-

Column Name	Data Type
Student_ID	INT
Name	VARCHAR(50)
Department	VARCHAR(30)
Age	INT

This table structure is called the **schema** because it defines how the data will be stored.

2. Instance

Definition

An **Instance** is the actual data stored in the database at a particular time. The instance changes whenever new records are inserted, updated, or deleted.

Features

- Contains real data.
- Changes frequently.
- Represents the current state of the database.

Example

Using the above Student schema, the instance may be:

Student_ID	Name	Department	Age
101	jaya	CSE	20
102	Priya	ECE	21
103	Arjun	IT	22

If another student is added, the instance changes, but the schema remains the same.

3. Logical Schema

Definition

The **Logical Schema** describes the logical structure of the database. It specifies what data is stored, the relationships among tables, and constraints, without considering how the data is physically stored.

Example

Student (Student_ID, Name, Department, Age)
Course (Course_ID, Course_Name)
Enrollment (Student_ID, Course_ID)

Here, only the database structure and relationships are defined.

Features

- Defines tables and attributes.
- Specifies relationships among tables.
- Independent of physical storage.
- Used by database designers.

4. Physical Schema

Definition:
A **Physical Schema** describes **how data is physically stored** in the database. It includes storage details such as files, indexes, and disk blocks.

Features:

- Describes physical storage of data.
- Includes files and indexes.
- Improves data retrieval speed.
- Managed by the DBMS.

Example:

Student_ID	Name	Department
101	jaya	CSE
102	Priya	ECE

Physical Storage:

- File: student.dat
- Index: B+ Tree on **Student_ID**
- Stored in disk blocks on the hard disk.

5. View Schema

Definition:
A View Schema (External Schema) is the part of the database that is visible to a specific user. It shows only the required data and hides unnecessary or confidential information.

Features:

- Shows only required data.
- Hides sensitive information.
- Improves security.
- Different users can have different views.

Example:

Student_ID	Name	Marks	Attendance
101	jaya	85	90%

Difference Between Logical Schema, Physical Schema, and View Schema

Logical Schema	Physical Schema	View Schema
Defines the logical structure of the database.	Defines how data is physically stored.	Defines what data a specific user can access.

Contains tables, attributes, and relationships.	Contains storage details, files, and indexes.	Contains only selected data.
Independent of storage devices.	Depends on storage devices.	Depends on user requirements.
Used by database designers.	Used by DBMS and DBAs.	Used by end users and applications.
Example: Student(Student_ID, Name, Age).	Example: Records stored using B+ Tree index.	Example: Student sees only Name and Department.

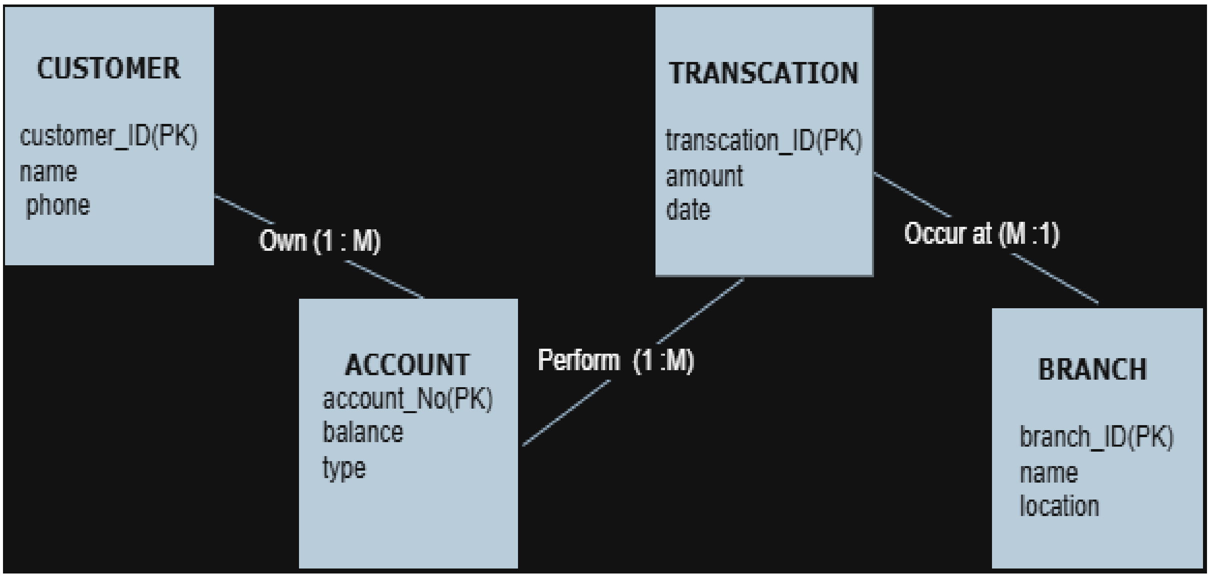
Conclusion

A **schema** defines the structure of a database, while an **instance** represents the actual data stored at a particular time. The **Logical Schema** defines the database structure, the **Physical Schema** describes how data is stored internally, and the **View Schema** provides customized views for different users. Together, these schemas provide data abstraction, security, and efficient database management.

Question 2

Draw an ER Diagram for a Bank Management System. Identify entities, attributes, relationships, and cardinality.

ER Diagram



Introduction

A **Bank Management System** stores and manages information about customers, accounts, loans, branches, employees, and transactions. An **Entity-Relationship (ER) Diagram** represents the entities, their attributes, relationships, and cardinality.

1. Entities and Attributes

Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone

Account

- Account_No (Primary Key)
- Account_Type
- Balance

Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

Loan

- Loan_ID (Primary Key)
- Loan_Amount
- Interest_Rate

Employee

- Employee_ID (Primary Key)
- Name
- Designation

Transaction

- Transaction_ID (Primary Key)
- Date
- Amount
- Type

2. Relationships

- Customer **owns** Account.
- Branch **maintains** Account.
- Customer **applies for** Loan.
- Employee **works at** Branch.
- Account **performs** Transaction.

3. Cardinality

Relationship	Description	Cardinality
Customer – Account	A customer can own one or more bank accounts.	1 : M (One-to-Many)
Branch – Account	A branch maintains multiple accounts.	1 : M (One-to-Many)
Customer – Loan	A customer can apply for one or more loans.	1 : M (One-to-Many)
Branch – Employee	A branch has many employees working in it.	1 : M (One-to-Many)
Account – Transaction	An account can have many transactions.	1 : M (One-to-Many)

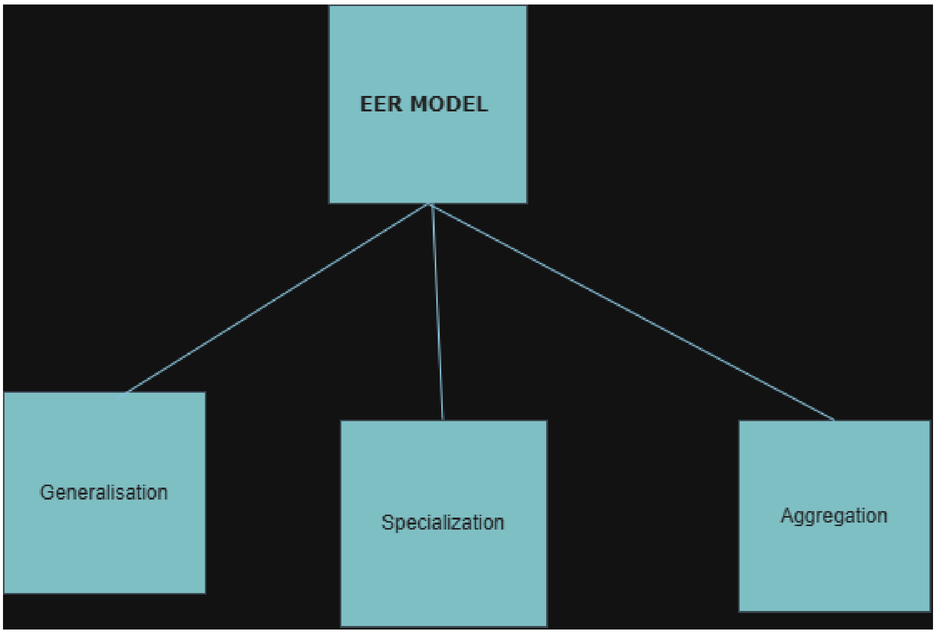
Conclusion

The ER Diagram of a Bank Management System includes Customer, Account, Branch, Loan, Employee, and Transaction as the main entities. Their relationships and cardinalities help organize the database efficiently, ensuring proper management of banking operations.

Question 3

Explain the Extended Entity-Relationship (EER) Model. Discuss the concepts of Generalization, Specialization, and Aggregation with examples.

Concept map



Introduction

The Extended Entity-Relationship (EER) Model is an advanced version of the Entity-Relationship (ER) Model. It provides additional features such as Generalization, Specialization, and Aggregation to represent complex real-world applications more effectively.

1. Generalization

Definition

Generalization is the process of combining two or more similar lower-level entities into a single higher-level entity based on their common attributes.

Features

- Combines similar entities.
- Creates a common parent entity.
- Reduces data redundancy.

Example

```
Savings Account
     \
      \
    Account
      /
Current Account
```

Here, Savings Account and Current Account are generalized into the Account entity.

2. Specialization

Definition

Specialization is the process of dividing one higher-level entity into multiple lower-level entities based on specific characteristics.

Features

- Divides a parent entity into child entities.
- Child entities inherit attributes from the parent.
- Adds specialized attributes.

Example

```
      Employee
      /    \
  Manager   Clerk
```

Here, Employee is specialized into Manager and Clerk.

3. Aggregation

Definition

Aggregation is a process in which a relationship between entities is treated as a higher-level entity so that it can participate in another relationship.

Features

- Treats a relationship as an entity.
- Represents complex relationships.
- Simplifies database design.

Example

```
      Customer
          |
       Places
         |
       Order    -------- Product
```

Here, the Order relationship is treated as a higher-level entity connected to Product.

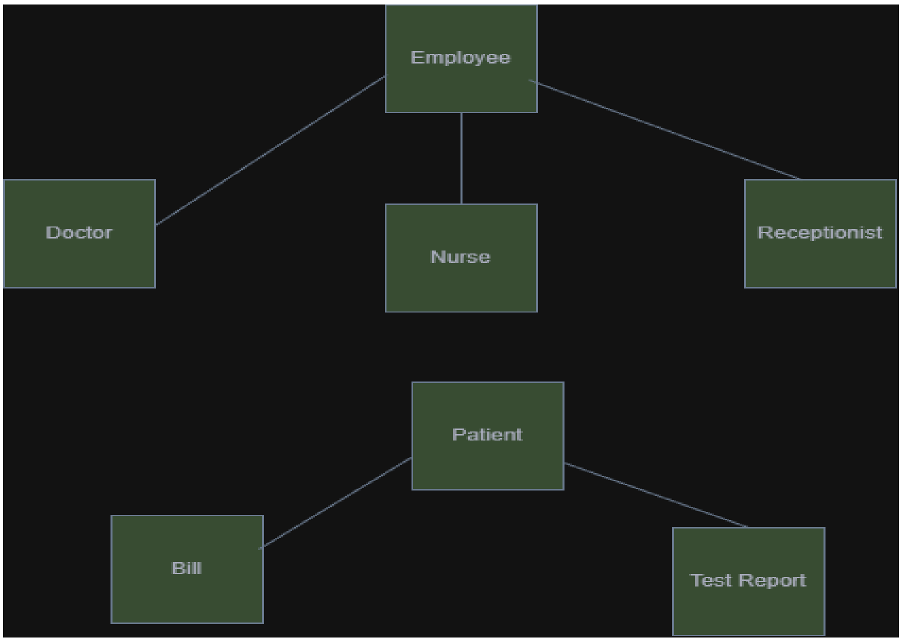
Conclusion

The EER Model extends the ER model by supporting Generalization, Specialization, and Aggregation. These concepts help represent complex database structures more accurately and improve database design.

Question 4

Design an EER Diagram for a Hospital Management System incorporating inheritance and weak entities.

Hospital Management System EER Diagram



Introduction

A Hospital Management System (HMS) is used to manage information about patients, doctors, nurses, rooms, bills, records, and test reports. The Extended Entity Relationship (EER) model represents these entities, their relationships, inheritance, and weak entities.

Explanation of the Hospital Management System EER Diagram

The Hospital Management System EER diagram represents the entities, attributes, relationships, inheritance, and weak entities involved in managing hospital operations.

1. Patient Entity

The **Patient** entity stores patient details such as **P-ID, Name, Age, Gender, Date of Birth (DOB), and Mobile Number**. A patient can consult doctors, pay bills, have test reports, and be assigned to a room.

2. Employee Entity (Super Entity)

The **Employee** entity is the **superclass** and contains common attributes like **E-ID, Name, Salary, Address, Mobile Number, and Sex**.

3. Inheritance (Specialization)

The **Employee** entity is specialized into three subclasses:

- **Doctor** – Attributes: Department, Qualification
- **Nurse**
- **Receptionist**

These subclasses inherit the common attributes of the **Employee** entity.

4. Room Entity

The **Room** entity contains **R-ID, Type, Capacity, and Availability**. Patients are assigned to rooms, and nurses govern the rooms.

5. Bills Entity

The **Bills** entity stores billing information such as **B-ID, Amount, and P-ID**. Each patient pays one or more bills.

6. Test Report (Weak Entity)

The **Test Report** stores **Test Type, Result, P-ID, and R-ID**. It depends on the **Patient** entity for its existence, making it a **weak entity**.

7. Records Entity

The **Records** entity stores **Record Number** and **Appointment Number**. These records are maintained by the receptionist.

8. Relationships

- Patient consults Doctor (M:N)
- Patient pays Bills (1:N)
- Patient has Test Reports (1:N)
- Patient is assigned to a Room (N:1)
- Nurse governs Rooms (M:M)
- Receptionist maintains Records (M:M)

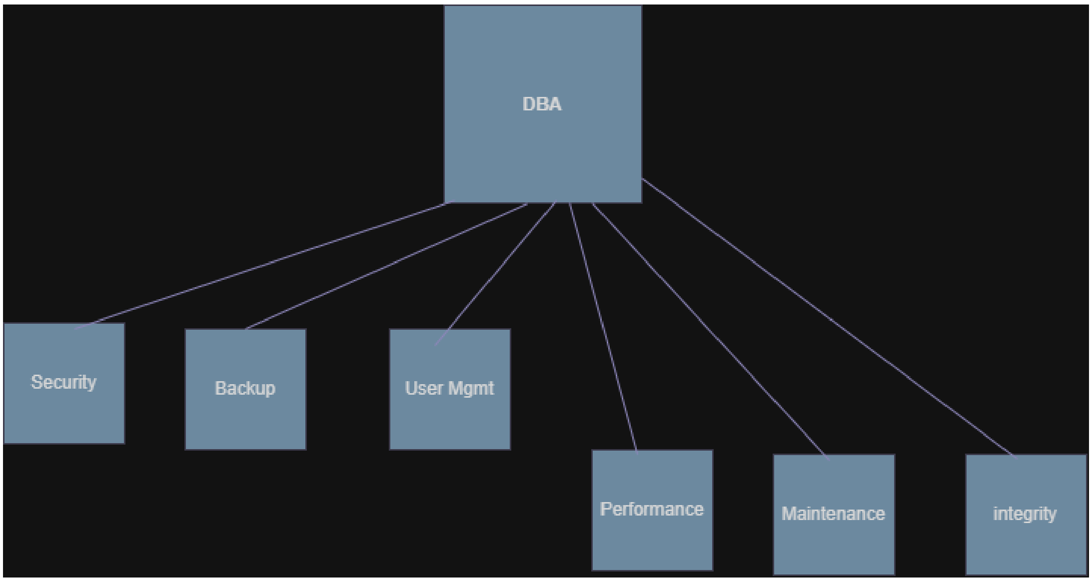
Conclusion

This EER diagram models the hospital database efficiently by using **inheritance** (Employee → Doctor, Nurse, Receptionist) and a **weak entity** (Test Report). It helps organize patient, employee, room, billing, and medical record information effectively.

Question 5

Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Concept Map



Introduction

A **Database Administrator (DBA)** is a person responsible for managing, maintaining, and securing a database. The DBA ensures that the database operates efficiently, remains secure, and is available to authorized users.

Definition

A **Database Administrator (DBA)** is a professional who installs, configures, maintains, and monitors databases to ensure data is stored securely and can be accessed efficiently.

Role of a DBA

- Manages the database system.
- Ensures database security.
- Maintains database performance.
- Performs backup and recovery.

- Controls user access and permissions.

Responsibilities of a DBA

1. **Database Installation and Configuration**
    - Installs and configures DBMS software.
2. **Database Design**
    - Creates database structures, tables, and relationships.
3. **Security Management**
    - Protects data from unauthorized access.
4. **User Management**
    - Creates user accounts and assigns permissions.
5. **Backup and Recovery**
    - Performs regular backups and restores data after failures.
6. **Performance Monitoring**
    - Monitors and improves database performance.

1. **Data Integrity**
    - Ensures data is accurate, consistent, and reliable.

7. **Database Maintenance**
    - Updates, patches, and maintains the database.

Advantages of a DBA

- Ensures data security.
- Improves database performance.
- Prevents data loss through backups.
- Maintains data consistency and availability.

Conclusion

A **Database Administrator (DBA)** plays a vital role in managing and maintaining databases. The DBA is responsible for database security, backup, recovery, performance optimization, and user management, ensuring that the database remains reliable, secure, and efficient.